

SELENA K. STEINBERG

Curriculum vitae

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EDUCATION

- Indiana University, School of Education**, Bloomington, IN | expected 2026
 Doctoral candidate in Learning Sciences
 Inquiry Methodologies Minor
 Advisor: Joshua Danish
 GPA 4.0
- Vanderbilt University, Peabody College**, Nashville, TN | 2021
 Master of Education Degree
 Child Studies- Clinical and Developmental Research Track
 Thesis: *Coding Choreography: Understanding Student Responses to Incompatibilities between Possibilities in Programming and Dance*
 GPA 4.0
- Tufts University**, Medford, MA | 2018
 Bachelor of Science Degree, Summa Cum Laude
 Child Study & Human Development Major, Mathematics Major
 Dance Minor, Drama Minor
 Senior Dance Capstone: *The Integration of Mathematics and Creative Dance*
 GPA 3.9

RESEARCH AGENDA

Theories: sociocultural theories, socio-critical theories, embodied cognition
Methods: interaction analysis, thematic analysis, discourse analysis (including mediated discourse analysis and discursive psychology), qualitative memos and coding
Topics: disciplinary identity in science/math, youth agency, performing arts/STEM integration

HONORS AND AWARDS

- ◆ **IU Center for Research on Learning and Technology Travel Scholarship** | 2025
- ◆ **IU Center for Research on Learning and Technology Travel Scholarship** | 2024
- ◆ **Runner Up Best Student Paper Award SIG-LS/ATL** | 2024
American Educational Research Association
along with Mengxi Zhou, Nitasha Mathayyas, Joshua Danish and Morgan Vickery
- ◆ **First Place Outstanding Research Award & \$200 Scholarship** | 2023
Indiana University Women's Research Poster Competition
along with Mengxi Zhou and Christina Stiso
- ◆ **IU Center for Research on Learning and Technology Travel Scholarship** | 2023

- ◆ **Indiana University Dean's Fellowship** | 2021-2025
- ◆ **Child Studies M.Ed. Excellence Award** | 2021
Presented by the faculty of the department to the graduating professional student from the Clinical and Developmental Research track who best exemplifies the goals of the program.
- ◆ **Best Paper** | 2020
International Conference of the Learning Sciences along with Corey Brady, Melissa Gresalfi, and Madison Knowe
- ◆ **Peabody Honors Scholarship** | 2019-2021
- ◆ **Summa Cum Laude** | 2018
- ◆ **Phi Beta Kappa Society** | 2018
- ◆ **Tufts University Alumni Association Senior Award** | 2018
Presented to six to twelve graduating members of the senior class for academic achievement, participation in campus and community activities, and leadership.
- ◆ **Honos Civicus Society** | 2018
Recognizes graduating students who engaged in meaningful service and leadership activities and strengthened civic life on campus and beyond during their time at Tufts.

PEER REVIEWED PUBLICATIONS

- Zhou, M., **Steinberg, S.**, Stiso, S., Danish, J., & Craig, K. (2023). Using network visualizations to engage elementary students in locally relevant data literacy. *Information and Learning Sciences*. 125(3/4), 209-231.
- Danish, J., Anton, G., Mathayas, N., Jen, T., Vickery, M., Lee, S., Tu, X., Cosic, L., Zhou, M., Ayalon, E., **Steinberg, S.**, Enyedy, N., & Ryan, Z. (2022). Designing for shifting learning activities. *The Journal of Applied Instructional Design*, 11(4), 169-185.
- Steinberg, S.**, Gresalfi, M., Vogelstein, L., Brady, C. (2022). Coding choreography: Understanding student responses to representational incompatibilities between dance and programming. *Journal of Research on Technology in Education*. 56(3), 314–331.
- Steinberg, S.**, Liu, T., Lense, M. (2021). Musical engagement and parent-child attachment in families with young children during the Covid-19 pandemic. *Frontiers in Psychology*. 10.3389/fpsyg.2021.641733
- Steinberg, S.**, Shivers, C.M., Liu, T., Cirelli, L.K., & Lense, M.D. (2021). Survey of the home musical environment of children with various developmental profiles. *Journal of Applied Developmental Psychology*, 75, 101296.

Under Review

- Steinberg, S.**, Lau, N.G., Cruz-Gonzalez, C., Coenraad, M., & Danish, J. Sharing and Connecting via Network Visualizations: An Asset-Based Approach to History Learning. [manuscript submitted for publication]
- Steinberg, S.** & Danish, J. “It Actually Made Me Want to Participate More”: Collective Embodied Modeling and Science Identity. [manuscript submitted for publication]
- Steinberg, S.**, Zou, X., Humburg, M., Hmelo-Silver, C., Danish, J., Glazewski, K., Rowe, J., & Lester, J. Exploring approaches to measuring student engagement across computer-supported collaborative learning contexts. [manuscript submitted for publication]
- Mathayas, N., Danish, J., Zhou, M., **Steinberg, S.**, Vickery, M., & Enyedy, N. Leveraging embodied, code-based, and diagrammatic models to foster fifth-grade students’ comprehension of ecosystem models and metamodeling. [manuscript submitted for publication]
- Greenberg, D., Quick, J., **Steinberg, S.**, Cruz Gonzalaz, C., Lau, N.G., Wolfe, B., & Fox, S. Developing a participatory design partnership. [manuscript submitted for publication]

PUBLISHED CONFERENCE PROCEEDINGS

- Steinberg, S.**, Danish, J., Mathayas, N., & Zhou, M. (2026). Many ways of “being” a beaver: Students’ perceptions of technology-enhanced and dramatized embodiment in science. International Society of the Learning Sciences.
- Steinberg, S.**, Lau, N.G., Cruz-Gonzalez, C., Danish, J., & Coenraad, M. (2026). Understanding history through data: Inviting and supporting student personal negotiation of historical data. International Society of the Learning Sciences.
- Lau, N.G., Stiso, C., **Steinberg, S.**, Zhou, M., Cruz-Gonzalez, C., Coenraad, M. & Danish, J. (2026). “Wait! Why is everyone connected to mine?!”: Contextualizing rule breaking as a design affordance. International Society of the Learning Sciences.
- Steinberg, S.**, Coenraad, M., Cruz-Gonzalez, C., Danish, J., & Lau, N. G. (2025). Connecting identity and classroom content: The role of network visualization in eliciting history students’ funds of identity. In Rajala, A., Cortez, A., Hofmann, R., Jornet, A., Lotz-Sisitka, H., & Markauskaite, L. (Eds.), Proceedings of the 19th International Conference of the Learning Sciences - ICLS 2025 (pp. 173-181). International Society of the Learning Sciences.
- Steinberg, S.** & Danish, J. (2025). Taking a scientist perspective: Constructing science identities through embodied modeling. In Rajala, A., Cortez, A., Hofmann, R., Jornet, A., Lotz-Sisitka, H., & Markauskaite, L. (Eds.), Proceedings of the 19th International Conference

- of the Learning Sciences - ICLS 2025 (pp. 1424-1428). International Society of the Learning Sciences.
- Cruz-Gonzalez, C., Coenraad, M., **Steinberg, S.**, Danish, J., Lau, N. G., & Hmelo-Silver, C. E. (2025). Interest vs. identity: Navigating methodological challenges to coding learners' funds of identity. In Rajala, A., Cortez, A., Hofmann, R., Jornet, A., Lotz-Sisitka, H., & Markauskaite, L. (Eds.), Proceedings of the 19th International Conference of the Learning Sciences - ICLS 2025 (pp. 2620-2622). International Society of the Learning Sciences.
- Zhou, M., **Steinberg, S.**, Craig, K., Danish, J., Wallace, C., Dunbar, K. M., Coenraad, M., & Hmelo-Silver, C. E. (2025). Exploring students' perceived affordances of physical and digital visualizations of data in Greek trade lessons. In Rajala, A., Cortez, A., Hofmann, R., Jornet, A., Lotz-Sisitka, H., & Markauskaite, L. (Eds.), Proceedings of the 19th International Conference of the Learning Sciences - ICLS 2025 (pp. 1439-1443). International Society of the Learning Sciences.
- Zhou, M., Danish, J., Fonteles, J. H., Davalos, E., Ayalon, E., Lane, A.C., S., A.T., Srivastava, N., **Steinberg, S.**, Anton, G., Enyedy, N., & Biswas, G. (2025). Interaction analysis practice within computational trends: Promises and challenges. In Rajala, A., Cortez, A., Hofmann, R., Jornet, A., Lotz-Sisitka, H., & Markauskaite, L. (Eds.), Proceedings of the 19th International Conference of the Learning Sciences - ICLS 2025 (pp. 2644-2646). International Society of the Learning Sciences.
- Fonteles, J., Davalos Anala, E., T S, A., Zhang, Y., Zhou, M., Ayalon, E., Lane, A., **Steinberg, S.**, Danish, J., Enyedy, N., & Biswas, G. (2024). A first step to using machine learning methods to enhance interaction analysis in embodied learning environments. In Proceedings of the 25th International Conference on Artificial Intelligence in Education (AIED 2024).
- Steinberg, S.** & Danish, J. (2024). "We were doing science, not just talking about science": Embodied learning and science identity development. In Lindgren, R., Asino, T. I., Kyza, E. A., Looi, C. K., Keifert, D. T., & Suárez, E. (Eds.), Proceedings of the 18th International Conference of the Learning Sciences - ICLS 2024 (pp. 306-313). International Society of the Learning Sciences.
- Steinberg, S.**, Quick, J., Cruz-Gonzalez, C., Greenberg, D., Lau, N. G., Fox, S., & Wolfe, B. (2024). Identity, power, and imagination at a children's museum. In Lindgren, R., Asino, T. I., Kyza, E. A., Looi, C. K., Keifert, D. T., & Suárez, E. (Eds.), Proceedings of the 18th International Conference of the Learning Sciences - ICLS 2024 (pp. 2075-2076). International Society of the Learning Sciences.
- Zhou, M., **Steinberg, S.**, Stiso, C., Danish, J., & Cruz-Gonzalez, C. (2024). Embodied and digital network visualization of data design for elementary students. In Lindgren, R., Asino, T. I., Kyza, E. A., Looi, C. K., Keifert, D. T., & Suárez, E. (Eds.), Proceedings of

- the 18th International Conference of the Learning Sciences - ICLS 2024 (pp. 1107-1110). International Society of the Learning Sciences.
- Danish, J., Mathayas, N., Zhou, M., **Steinberg, S.**, & Vickery, M. (2024). Character based models and computational and embodied action tweaking for sensemaking. In Clarke-Midura, J., Kollar, I., Gu, X., & D'Angelo, C. (Eds.), *Proceedings of the 17th International Conference on Computer-Supported Collaborative Learning - CSCL 2024* (pp. 391-392). International Society of the Learning Sciences.
- Zhou, M., Fonteles, J., Danish, J., Davalos, E., **Steinberg, S.**, Biswas, G., & Enyedy, N. (2024). Exploring artificial intelligence supported interaction analysis. In Lindgren, R., Asino, T. I., Kyza, E. A., Looi, C. K., Keifert, D. T., & Suárez, E. (Eds.), *Proceedings of the 18th International Conference of the Learning Sciences - ICLS 2024* (pp. 2327-2328). International Society of the Learning Sciences.
- Vickery, M., Mathayas, N., **Steinberg, S.**, & Humburg, M. (2024). From reactive to proactive: Considering socio-affective experiences in inclusive embodied activity design. In Tancredi, S. & Vickery, M. (Co-Chairs), *Learning for every body: Intersectional dimensions of embodied learning [Symposium]*. In Lindgren, R., Asino, T. I., Kyza, E. A., Looi, C. K., Keifert, D. T., & Suárez, E. (Eds.), *Proceedings of the 18th International Conference of the Learning Sciences - ICLS 2024*. (pp. 2037-2044). Buffalo, USA: International Society of the Learning Sciences.
- Steinberg, S.**, Hmelo-Silver, C., Zou, X., Danish, J., & Lester, J. (2023). Seeing student engagement in classroom video: Affordances of cognitive and sociocultural frameworks. In Damşa, C., Borge, M., Koh, E., & Worsley, M. (Eds.), *Proceedings of the 16th International Conference on Computer-Supported Collaborative Learning-CSCL 2023*, pp. 123-130. International Society of the Learning Sciences.
- Steinberg, S.**, Zhou, M., Vickery, M., Mathayas, N., & Danish, J. (2023). Making sense of modes in collective embodied science activities. In Blikstein, P., Van Aalst, J., Kizito, R., & Brennan, K. (Eds.), *Proceedings of the 17th International Conference of the Learning Sciences-ICLS 2023*, pp. 1218-1221. International Society of the Learning Sciences.
- Zhou, M., **Steinberg, S.**, Stiso, C., Danish, J. A. & Craig, K. (2023). Exploring network visualization of data in elementary classroom. In Blikstein, P., Van Aalst, J., Kizito, R., & Brennan, K. (Eds.), *Proceedings of the 17th International Conference of the Learning Sciences - ICLS 2023* (pp. 465-472). International Society of the Learning Sciences.
- Stiso, C., Zhou, M., **Steinberg, S.**, Danish, J. A. & Craig, K. (2023). Visualizing funds of identity: Using network software to model collective identity in a classroom. In Blikstein, P., Van Aalst, J., Kizito, R., & Brennan, K. (Eds.), *Proceedings of the 17th International Conference of the Learning Sciences-ICLS 2023*, pp. 1919-1920. International Society of the Learning Sciences.

- Steinberg, S.** & Danish, J. (2022). Balancing agency and accountability to support learning in playful, embodied science activities. In Chinn, C., Tan, E., Chan, C., & Kali, Y. (Eds.), *Proceedings of the 16th International Conference of the Learning Sciences - ICLS 2022* (pp. 1149-1152). International Society of the Learning Sciences.
- Steinberg, S.**, & Gresalfi, M.S. (2021). Agency and expressivity in programming play. In Vries, E., Hod, Y., & Ahn J. (Eds.). *Proceedings of the 15th International Conference of the Learning Sciences*, pp. 581-584. Bochum, Germany: International Society of the Learning Sciences.
- Gresalfi, M.S., Brady, C., Knowe, M., **Steinberg, S.** (2020). Engaging in a new practice: What are students doing when they are “doing” debugging? In M.S. Gresalfi & I.S. Horn (Eds.). *The Interdisciplinarity of the Learning Sciences*, 14th International Conference of the Learning Science, pp. 199-206. Nashville, Tennessee: International Society of the Learning Sciences.
- Brady, C., Gresalfi, M.S., **Steinberg, S.**, & Knowe, M. (2020). Debugging for art’s sake: Beginning programmers’ debugging activity in an expressive coding context. In M.S. Gresalfi & I.S. Horn (Eds.). *The Interdisciplinarity of the Learning Sciences*, 14th International Conference of the Learning Science, pp. 1229-1236. Nashville, Tennessee: International Society of the Learning Sciences. (Best Paper Award)
- Gresalfi, M., Brady, C., Knowe, M., & **Steinberg, S.** (2020). Framing the task: Debugging vs. exploring as opportunities to learn. In Kafai, Y. (Chair), *Turning bugs into learning opportunities: Understanding debugging processes, perspectives, and pedagogies [symposium]*. In M.S. Gresalfi & I.S. Horn (Eds.). *The Interdisciplinarity of the Learning Sciences*, 14th International Conference of the Learning Science, pp. 374-381. Nashville, Tennessee: International Society of the Learning Sciences.

REFEREED TALKS AND PRESENTATIONS

- Cruz-Gonzalez, C., Coenraad, M., **Steinberg, S.**, Zhou, M., & Danish, J. (2026, April). *Making meaning with data: Integrating data literacy in ELA*. Poster presented at the American Education Research Association (AERA) Annual Conference, Los Angeles, CA.
- Steinberg, S.**, Mathayas, N., Zhou, M., & Danish, J. (2025, April). *The role of embodiment in scientific modeling and learning about wetlands ecosystems*. Poster presented at the American Education Research Association (AERA) Annual Conference, Denver, CO.
- Zhou, M., **Steinberg, S.**, Craig, K., & Danish, J. (2025, April). *Students’ perceived affordances of three representations in Greek trade network activity*. Poster presented at the American Education Research Association (AERA) Annual Conference, Denver, CO.
- Zhou, M., **Steinberg, S.**, & Danish, J. (2025, February). *Embodiment and network visualization of data for early adolescents*. DSE-K12: Data Science Education K-12: Research to Practice Conference, San Antonio, TX.

Liu, T.*, **Steinberg, S.***, & Lense, M. (2024, July). *Children's agency and engagement during parent-child singing activities in autistic and non-autistic preschoolers*. Presented at the Society for Music Perception and Cognition (SMPC) 2024 conference. *co-first authors

Steinberg, S., Ćosić, L., Anton, G., Zhou, M., & Danish, J. (2024, April). *"Happy Juice!": Expansive notions of energy transfer through theatre in the science classroom*. Paper presented at the American Education Research Association (AERA) Annual Conference, Philadelphia, PA.

Zhou, M., **Steinberg, S.**, Danish, J., Stiso, C., Cruz-Conzalez, C. (2024, April). *Elementary students' reflection on embodied and digital data literacy representations*. Paper presented at the American Education Research Association (AERA) Annual Conference, Philadelphia, PA. (Best Student Paper Award Nomination).

Stiso, C., Zhou, M., **Steinberg, S.**, Danish, J., Craig, K. (2024, April). *Using network software to model collective identity in a classroom*. Paper presented at the American Education Research Association (AERA) Annual Conference, Philadelphia, PA.

Mathayas, N., Zhou, M., Danish, J., Vickery, M., **Steinberg, S.**, Ryan, Z., Tu, X., & Devine, I. (2024, April). *The role of embodied modeling on fifth grade students' perspectives on ecosystems thinking and metamodeling*. Paper presented at the American Education Research Association (AERA) Annual Conference, Philadelphia, PA.

Zhou, M., Mathayas, N., Danish, J., Vickery, M., & **Steinberg, S.** (2024, April). *Exploring students' divergent interpretations while studying ecosystems in an embodied mixed reality environment*. Poster presented at the American Education Research Association (AERA) Annual Conference, Philadelphia, PA. (AERA SIG Learning Sciences and Advanced Technology for Learning Runners up for Best Student Paper award).

Steinberg, S. (2023, October). *"I was able to understand it better than normal science": Students' identity work in embodied science activities*. Poster presented at the Learning Sciences Graduate Student Conference (LSGSC), Iowa City, IA.

Zhou, M., Stiso, C., **Steinberg, S.** (2023, October). *Exploring elementary students' negotiation process in network data creation*. Poster presented at the Learning Sciences Graduate Student Conference (LSGSC), Iowa City, IA.

Steinberg, S. & Danish, J. (2023, April). *The role of feedback in shaping student agency in technology-supported embodied science activities*. In Whose agency? What learning? Whence power? American Educational Research Association (AERA) Annual Meeting, Chicago, IL. <https://aera23-aera.ipostersessions.com/Default.aspx?s=3E-3C-0D-76-07-04-E6-E1-C0-FD-8B-39-22-A1-F2-FC>

Zhou, M., Stiso, C., & **Steinberg, S.** (2022, October). *Supporting students' data reasoning ability through embodied activities*. Poster presented at the Learning Sciences Graduate Student Conference (LSGSC), Bloomington, IN.

Ryan, Z., Tu, X., Vickery, M., Zhou, M., Humburg, M., **Steinberg, S.**, Lee, S., Jen, T., Cosic, L., (2022, October). Perspectives on play across the learning sciences: Discussing the past, present, and future aims of play and learning. In Proceedings of Learning Sciences Graduate Student Conference (LSGSC) 2022.

Gresalfi, M.S., **Steinberg, S.**, & Love, C. (2022, April). *From challenge to enjoyment: Designing to support identification with computer science*. In Sullivan, F.R. (Chair), Inclusion in computer science education: Documenting, designing, and supporting transformative learning in computer science classrooms [symposium]. Paper presented at the American Educational Research Association (AERA) Annual Conference, San Diego, CA.

INVITED TALKS AND OTHER PRESENTATIONS

Steinberg, S., & Zou, X. (2023, July 26). *Theorizing and operationalizing student engagement in CSCL contexts*. Talk given at the EngageAI Institute Brown Bag Series.

Zhou, M., Stiso, C., **Steinberg, S.** (2023). Mediating elementary students' data literacy through network visualization. 2023 Women's Research Poster Competition at Indiana University Bloomington. (Awarded first place outstanding research award for graduate students in the technology and computing category).

RESEARCH EXPERIENCE

- ◆ *NSF- DRL- 1908632*: Generalized Embodied Modeling to support Science through Technology Enhanced Play
- ◆ *NSF- DRL- 2241705*: Integrating Students' Interests, Identities and Ways of Knowing with Network Visualization Tools to Explore Data Literacy Concepts
- ◆ *NSF- DRL- 2112635*: AI Institute for Engaged Learning
- ◆ *NSF- DRL- 1742257*: Foregrounding Agency versus Structure as Models for Designing Integrated Mathematics and Computational Thinking Curriculum
- ◆ *1844332-38-C-18NEA*: Social-Emotional Development and Community Building through Music
- ◆ *N/A*: Participatory Design at a Children's Museum: Co-Designing for Equitable Out-of-School Learning
- ◆ *Dissertation*: Imagining a More Embodied and Artful Math and Science Education

TEACHING & RELATED EXPERIENCE

Instructor of Record, Indiana University
Course: EDUC-I 251 Learning Theories for Teachers
Course: EDUC-M 101 Field Experience

| **Fall 2024, Spring 2025**

Roles: Designed, taught, and graded curriculum covering major learning theories (behaviorism, cognitive theories, sociocultural theories, constructivism, and critical theories) and their application to the classroom for undergraduate preservice teachers

Teaching Assistant, Vanderbilt University | **Spring 2021**

Course: Introduction to Statistical Analysis

Professor: Dr. Maria Osina

Roles: held office hours; graded assignments and exams; communicated with class regarding assignments and course announcements

Teaching Assistant, Vanderbilt University | **Spring 2020**

Course: Introduction to Statistical Analysis

Professor: Dr. Joseph Rodgers

Roles: planned and taught mini-lab sessions; held office hours; graded assignments, quizzes, and papers; communicated with class regarding assignments and course announcements

Interpreter, National Museum of Mathematics | **2019**

The National Museum of Mathematics is an interactive museum in New York City that aims to make math engaging for all ages.

Roles: welcomed, engaged, and assisted families, school groups, and other visitors as they explored the museum; answered questions and explained the mathematics behind the exhibits

Research, Policy & Collective Action Intern, Theatre Communications Group | **2019**

Member of a three-person team focusing on arts advocacy, research, and arts education in the not-for-profit professional theatre field.

Roles: collected and organized information and data; edited online resources; created “Field at a Glance” document; assisted in designing and refining survey questions

Program Assistant, Johns Hopkins Center for Talented Youth | **2018**

Courses: Geometry, Robotics

Roles: assisted students in grades 4-7 with course material one-on-one and in small groups; planned and supervised daily recreational activities; assisted instructor with teaching, lesson planning, and paperwork

Teaching Assistant, Johns Hopkins Center for Talented Youth | **2017**

Course: Mathematical Modeling

Roles: assisted students in grades 7-11 with course material one-on-one and in small groups; developed lesson plans for and taught one class session every day; graded assignments and pre/post exams; assisted with student evaluations

PROFESSIONAL SERVICE

◆ **President** | **2023-2024**

Indiana University Learning Sciences Graduate Student Association

- ◆ **Events and Communications Officer** | **2022-2023**
Indiana University Learning Sciences Graduate Student Association
- ◆ **Reviewer**
 - Instructional Science | **2025**
 - International Society of the Learning Sciences | **2023-2026**
 - Education Inquiry | **2023**
 - Information and Learning Sciences | **2023**
- ◆ **Tufts University Alumni Interviewer** | **2018-2022**

PROFESSIONAL MEMBERSHIPS

- ◆ International Society of the Learning Sciences | **2021-present**
- ◆ American Educational Research Association | **2022-present**
 Learning Sciences SIG | **2024-present**